



BAULKHAM HILLS HIGH SCHOOL

2014

YEAR 12 HSC ASSESSMENT TASK 1

Mathematics Extension 1

General Instructions

- Reading time – 5 minutes
- Working time – 55 minutes
- Write using black or blue pen
- Board-approved calculators may be used
- All necessary working should be shown in every question
- Marks may be deducted for careless or badly arranged work
- Attempt all questions

Total marks – 36

This paper consists of 3 questions

Questions 1-3 – Extended Response 36 marks

Start a new page for each question.

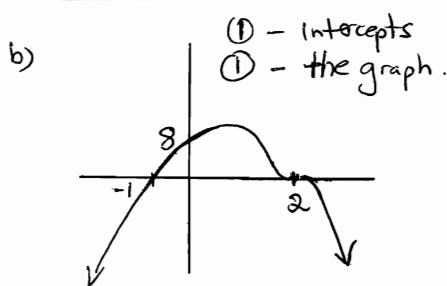
Question 1 (17 marks) - Start a new page		Marks
a)	A polynomial $P(x) = ax^n(x - 1)^2 + 5x + 7$ is monic of degree 6. (i) Find the value of a. (ii) Find the value of n. (iii) What is the remainder when $P(x)$ is divided by $(x - 1)$?	1 1 1
b)	Sketch the graph of $y = (2 - x)^3(x + 1)$ showing the x and y intercepts.	2
c)	Solve $x^3 + 3x^2 - 9x - 27 = 0$	3
d)	If α, β and γ are the roots of $2x^3 - 3x + 4x + 2 = 0$, find:- (i) $\alpha + \beta + \gamma$ (ii) $\alpha\beta\gamma$ (iii) $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$ (iv) $(\alpha - 1)(\beta - 1)(\gamma - 1)$	1 1 2 2
e)	The polynomial $P(x) = x^3 + ax^2 + bx + 4$ has $x + 2$ as a factor. When $P(x)$ is divided by $x - 1$ the remainder is 9. Find the value of a and b.	3

Question 2 (9 marks) - Start a new page		
a)	Use mathematical induction to prove $\frac{1}{1.2} + \frac{1}{2.3} + \dots + \frac{1}{n(n+1)} = \frac{n}{n+1}$ where n is a positive integer.	3
b)	Two roots of the polynomial $2x^3 + x^2 - kx + 6 = 0$ are equal in absolute value but opposite in sign. Find the value of k .	3
c)	What is the remainder when $x^{99} - 99$ is divided by $x^2 - 1$?	3

Question 3 (10 marks) - Start a new page		Marks
a)	Use mathematical induction to prove $n^3 + 4n$ is divisible by 8 if n is an even integer, where $n \geq 2$.	4
b)	Two of the roots of the equation $x^3 + ax^2 + b = 0$ are reciprocals of each other.	
	(i) Show that the third root is equal to $-b$	1
	(ii) Show that $a = b - \frac{1}{b}$	2
	(iii) Show that the 2 roots which are reciprocals will be real if $-\frac{1}{2} \leq b \leq \frac{1}{2}$	3

	End of Examination	
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- a) (i) $a = 1$ ①
 (ii) $n = 4$ ①
 (iii) $P(1) = 12$ ①



c) $P(x) = x^3 + 3x^2 - 9x - 27$
 $P(3) = 27 + 27 - 27 - 27$
 $\therefore (x-3)$ is a factor. ①

$$\begin{array}{r} x^2 + 6x + 9 \\ x-3 \overline{) x^3 + 3x^2 - 9x - 27} \\ \underline{x^3 - 3x^2} \\ 6x^2 - 9x \\ \underline{6x^2 - 18x} \\ 9x - 27 \\ \underline{9x - 27} \\ 0 \end{array}$$

$\therefore P(x) = (x-3)(x+3)^2$ ①

$\therefore$ Solution $x = 3, -3$ ①

d) $2x^3 - 3x^2 + 4x + 2 = 0$
 (i) $\alpha + \beta + \gamma = \frac{3}{2}$ ①
 (ii) $\alpha\beta\gamma = -1$ ①
 (iii) $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = \frac{\beta\gamma + \alpha\gamma + \alpha\beta}{\alpha\beta\gamma}$
 $= \frac{2}{-1}$ ①
 $= -2$ ①

(iv) $(\alpha-1)(\beta-1)(\gamma-1)$
 $= (\alpha-1)(\beta\gamma - \beta - \gamma + 1)$
 $= \alpha\beta\gamma - \alpha\beta - \alpha\gamma + \alpha - \beta\gamma + \beta + \gamma - 1$
 $= \alpha\beta\gamma - (\alpha\beta + \alpha\gamma + \beta\gamma) + \alpha + \beta + \gamma - 1$
 $= -1 - 2 + \frac{3}{2} - 1$ ①
 $= -\frac{5}{2}$ ①

e) $P(x) = x^3 + ax^2 + bx + 4$
 $P(-2) = 0 \therefore -8 + 4a - 2b + 4 = 0$
 $4a - 2b = 4$
 $2a - b = 2$ -- ①
 $P(1) = 9 \quad 1 + a + b + 4 = 9$
 $a + b = 4$ -- ②

$2a - b = 2$ -- ①
 $a + b = 4$ -- ②

①+② $3a = 6 \rightarrow a = 2$
 $b = 2$

① mark for both eq's
 ② marks for solving for "a" and "b". (1 mark off for each error)

Question 2 [9]

a) Prove
 $\frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \dots + \frac{1}{n(n+1)} = \frac{n}{n+1}$

Step 1 Prove true for $n=1$

LHS = $\frac{1}{2}$ RHS = $\frac{1}{1+1} = \frac{1}{2}$

LHS = RHS $\therefore$ true for $n=1$.

(accept $\frac{1}{1 \cdot 2} = \frac{1}{1+1}$)

Step 2 Assume true for $n=k$

$\frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \dots + \frac{1}{k(k+1)} = \frac{k}{k+1}$ ①

Step 3. Prove true for $n=k+1$
 i.e.

$\frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \dots + \frac{1}{k(k+1)} + \frac{1}{(k+1)(k+2)}$
 $= \frac{k+1}{k+2}$

from Assumption ①

$\frac{k}{k+1} + \frac{1}{(k+1)(k+2)} = \frac{k+1}{k+2}$

LHS = $\frac{k(k+2) + 1}{(k+1)(k+2)}$

$= \frac{k^2 + 2k + 1}{(k+1)(k+2)}$

$= \frac{(k+1)^2}{(k+1)(k+2)}$

$= \frac{k+1}{k+2}$

= RHS

$\therefore$ Proven true for $n=k+1$
 if true for $n=k \therefore$ true
 for $n=1, n=2$ & for all
 n by mathematical
 induction.

① mark for steps 1, 2 and 4 correct

② marks for successfully proving step 3.

2b) $2x^3 + x^2 - kx + 6 = 0$

$\alpha, -\alpha, \beta$

$\therefore \alpha + -\alpha + \beta = -\frac{1}{2}$

$\beta = -\frac{1}{2}$ -- ①

$\alpha\beta - \alpha\beta - \alpha^2 = -\frac{k}{2}$

$\alpha^2 = \frac{k}{2}$ -- ②

$-\alpha^2\beta = -\frac{6}{2}$

$\alpha^2\beta = 3$ -- ③

sub ① into ③

$\alpha^2(-\frac{1}{2}) = 3$

$\alpha^2 = -6$

sub into ②

$-6 = \frac{k}{2}$

$k = -12$

Alternatively sub $\beta = -\frac{1}{2}$ back into equation & solve.

Marking scheme

① for the 3 equations

② marks for using eq's to solve for k .

c) $P(x) = x^{99} - 99$

$P(x) = (x^2 - 1)Q(x) + ax + b$

$\therefore x^{99} - 99 = (x^2 - 1)(Q(x)) + ax + b$

Sub in $x=1$

(1)⁹⁹ - 99 = $a(1) + b$

$a + b = -98$ -- ①

sub in $x=-1$

(-1)⁹⁹ - 99 = $-a + b$

$-a + b = -100$ -- ②

①+② $2b = -198$
 $b = -99$

$a = 1$

$\therefore$ remainder is $x - 99$

① for $(x^2 - 1)Q(x) + ax + b$

① for equations ① + ②

① for solving successfully, but must write remainder $x - 99$.

3a) Prove $n^3 + 4n$ is $\div$ by 8 if n is even

Step 1. Prove true for $n=2$

$$2^3 + 4(2) = 16$$

$$\frac{16}{8} = 2 \therefore \text{true for } n=2.$$

Step 2. Assume true for $n=k$

$$\text{i.e. } \frac{k^3 + 4k}{8} = M \text{ (where } M \text{ is an integer)}$$

$$\text{i.e. } k^3 = 8M - 4k. \quad \text{---(1)}$$

Step 3. Prove true for $n=k+2$.

$$\text{i.e. } (k+2)^3 + 4(k+2) \text{ is } \div \text{ by } 8, k \text{ even.}$$

$$k^3 + 6k^2 + 16k + 16 \text{ is } \div \text{ by } 8$$

sub in (1)

$$\begin{aligned} & 8M - 4k + 6k^2 + 16k + 16 \\ &= 8M + 6k^2 + 12k + 16 \quad \text{---(2)} \end{aligned}$$

but k is even $\therefore$
let $k=2a$ where a is a positive integer

$$\begin{aligned} &= 8M + 6(2a)^2 + 12(2a) + 16 \\ &= 8M + 24a^2 + 24a + 16 \end{aligned}$$

$$= 8(M + 3a^2 + 3a + 2)$$

$$\text{and since } 8 \text{ is a factor and } M \text{ and } a \text{ integers}$$

$(k+2)^3 + 4(k+2)$ is $\div$ by 8
step 4 proven true for $k+2$
if true for k $\therefore$
true for 2, 4, 6, ... $\therefore$
for all even n by
Mathematical Induction.

① mark for 1, 2 and 4

Step 3 - 3 marks

① for substituting $k+2$.

① for expanding $(k+2)^3$ correctly & getting to expression ②

① for substituting $k=2a$.

NB If don't state M is an integer lose 1 mark

NB If make same error in step 4 as in earlier M.I. question 2a) don't deduct marks twice.

$$b) x^3 + ax^2 + b = 0$$

roots $\alpha, \frac{1}{\alpha}, \beta$

$$(i) \alpha \times \frac{1}{\alpha} \times \beta = -b \therefore \beta = -b \quad \text{① mark}$$

$$(ii) \alpha + \frac{1}{\alpha} - b = -a \therefore \alpha + \frac{1}{\alpha} = b - a \quad \text{---(1)}$$

$$\alpha(\frac{1}{\alpha}) + \alpha(\beta) + \frac{\beta}{\alpha} = 0$$

$$\text{i.e. } 1 - \alpha b - \frac{b}{\alpha} = 0$$

$$1 - b(\alpha + \frac{1}{\alpha}) = 0 \quad \text{---(2)} \quad \text{① mark}$$

sub ① into ②

$$1 - b(b - a) = 0$$

$$-\frac{1}{b} = (b - a)$$

$$a = b - \frac{1}{b} \quad \text{---(3)} \quad \text{① mark}$$

sub in ③ $\alpha + \frac{1}{\alpha} = b - (b - \frac{1}{b})$

$$\alpha + \frac{1}{\alpha} = b - (b - \frac{1}{b})$$

$$\alpha + \frac{1}{\alpha} = \frac{1}{b}$$

$$\alpha^2 b - \alpha + b = 0 \quad \text{---(4)}$$

$$\therefore \alpha = \frac{1 \pm \sqrt{1 - 4b^2}}{2b}$$

roots are real if

$$1 - 4b^2 \geq 0$$

$$(1 + 2b)(1 - 2b) \geq 0$$



$\therefore$ roots are real

$$\text{if } -\frac{1}{2} \leq b \leq \frac{1}{2}$$

1 mark for sub. in ③ into

① & getting eq'n ④

1 mark for using quad. form. correctly.

1 mark for correct conclusion using Δ .